

# Mount Kenya University

## Unlocking Infinite Possibilities

**Name:** CHANDLER MATERE SIMIYU

**Admission Number:** BSCCS/2024/58571

**Unit Title:** ADVANCED WEB DESIGN & DEVELOPMENT

**Unit Code:** BIT3208

**Unit Name:** Advanced Web Design and Development

**Lecturer:** MR. NYORO MICHAEL

Year III, Semester II

**Project Title:** NoteKeeper - Full Stack Web Application

**Technologies Used:** [Node.js](#) + Express + MySQL

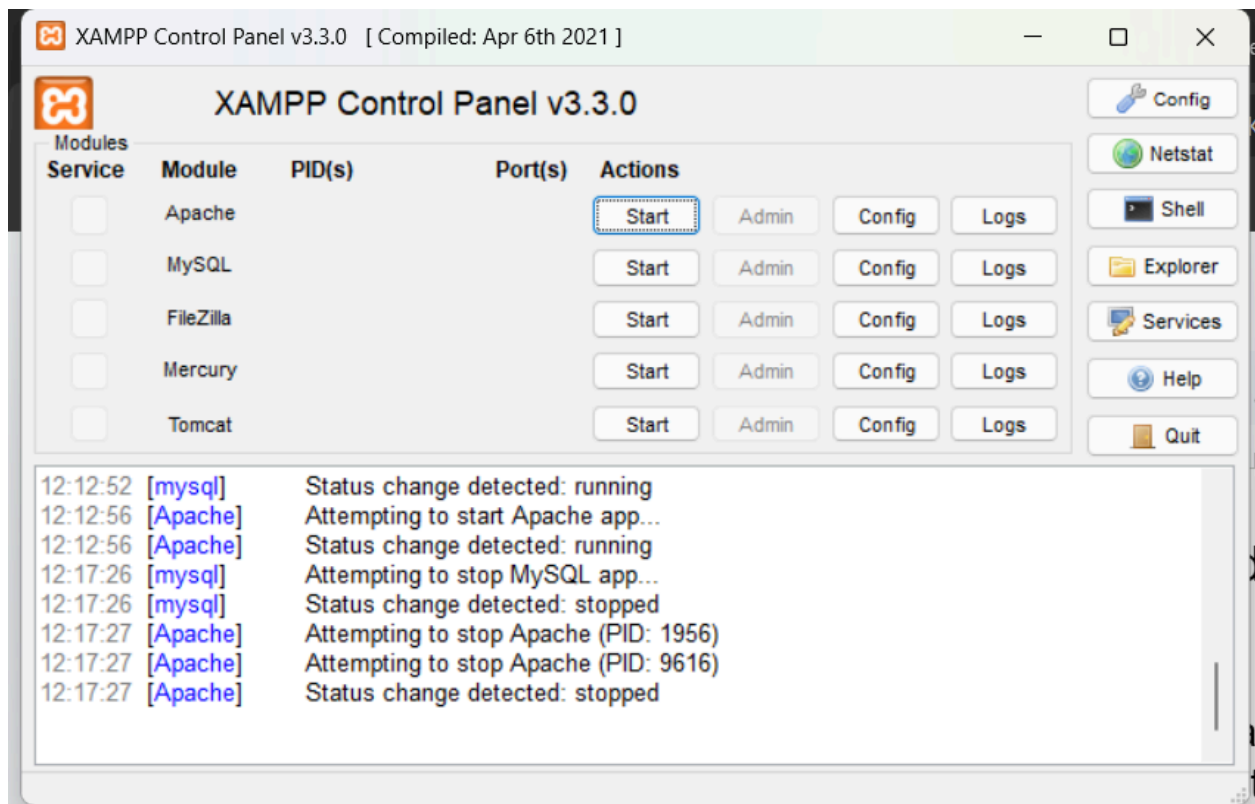
**GitHub Link:** <https://github.com/Simiyuu/notes-app>

|                                                                                                                    |           |
|--------------------------------------------------------------------------------------------------------------------|-----------|
| <b>Week 1: Local Environment Setup.....</b>                                                                        | <b>4</b>  |
| Fig 1: XAMPP Control Panel showing the local development environment installed on the machine.....                 | 4         |
| Fig 2: XAMPP Control Panel showing Apache and MySQL running on their default ports.....                            | 5         |
| Fig 3: Browser showing the XAMPP localhost welcome page.....                                                       | 5         |
| Fig 4: Hello World Test.....                                                                                       | 6         |
| Fig 5: Database Connection Test.....                                                                               | 6         |
| <b>Week 2: Wireframes and GUI Design.....</b>                                                                      | <b>7</b>  |
| Fig 1: Figma wireframe detailing the planned layout and user interface for the NoteKeeper login page.....          | 7         |
| Fig 2: Browser showing the Login page with username and password fields used to authenticate registered users..... | 8         |
| Fig 3: Mobile View Mockup.....                                                                                     | 9         |
| Fig 4: Color Theme and Navigation Structure.....                                                                   | 10        |
| <b>Week 3: JavaScript and PHP Basics.....</b>                                                                      | <b>11</b> |
| Fig1: Javascript Form Validation.....                                                                              | 11        |
| Fig 2: Password Strength Checker.....                                                                              | 12        |
| Fig 3: Shows the Express.js route syntax used in server.js to handle HTTP requests.....                            | 13        |
| Fig 4 shows the database connection configuration and initDB function in server.js.....                            | 14        |
| Fig 5: Dynamic User Input Handling.....                                                                            | 14        |
| <b>Week 4 Task Schedule.....</b>                                                                                   | <b>15</b> |
| Task 1 - Understanding Server-Side Programming.....                                                                | 15        |
| Fig 1: Simple Processing Page.....                                                                                 | 15        |
| Fig 2: Dynamic Welcome Page.....                                                                                   | 15        |
| Fig3: Form Submission System.....                                                                                  | 16        |
| Task 2 - HTML Forms and PHP Integration.....                                                                       | 17        |
| Fig 1: Registration Form.....                                                                                      | 17        |
| Fig 2: Login Form.....                                                                                             | 17        |
| Task 3 - Simple Authentication.....                                                                                | 18        |
| Fig 1: Basic Login Page.....                                                                                       | 18        |
| Fig 2: Username/passwords validation.....                                                                          | 19        |
| Fig 3: Session-based welcome page.....                                                                             | 20        |
| Task 4 - Backend Folder Organization.....                                                                          | 20        |
| Fig 1: Project Structure.....                                                                                      | 20        |
| Task 5 - Introduction to Modern Backend Alternatives.....                                                          | 21        |
| Fig : PHP Modern Alternative (Node.js).....                                                                        | 21        |
| <b>Week 5 - Advanced Web Design and Development.....</b>                                                           | <b>22</b> |
| Task 1: Database Creation.....                                                                                     | 22        |
| Fig 1: notes-app database.....                                                                                     | 22        |
| Task 2: Creating Tables.....                                                                                       | 22        |
| Fig 1: notes-app tables.....                                                                                       | 22        |

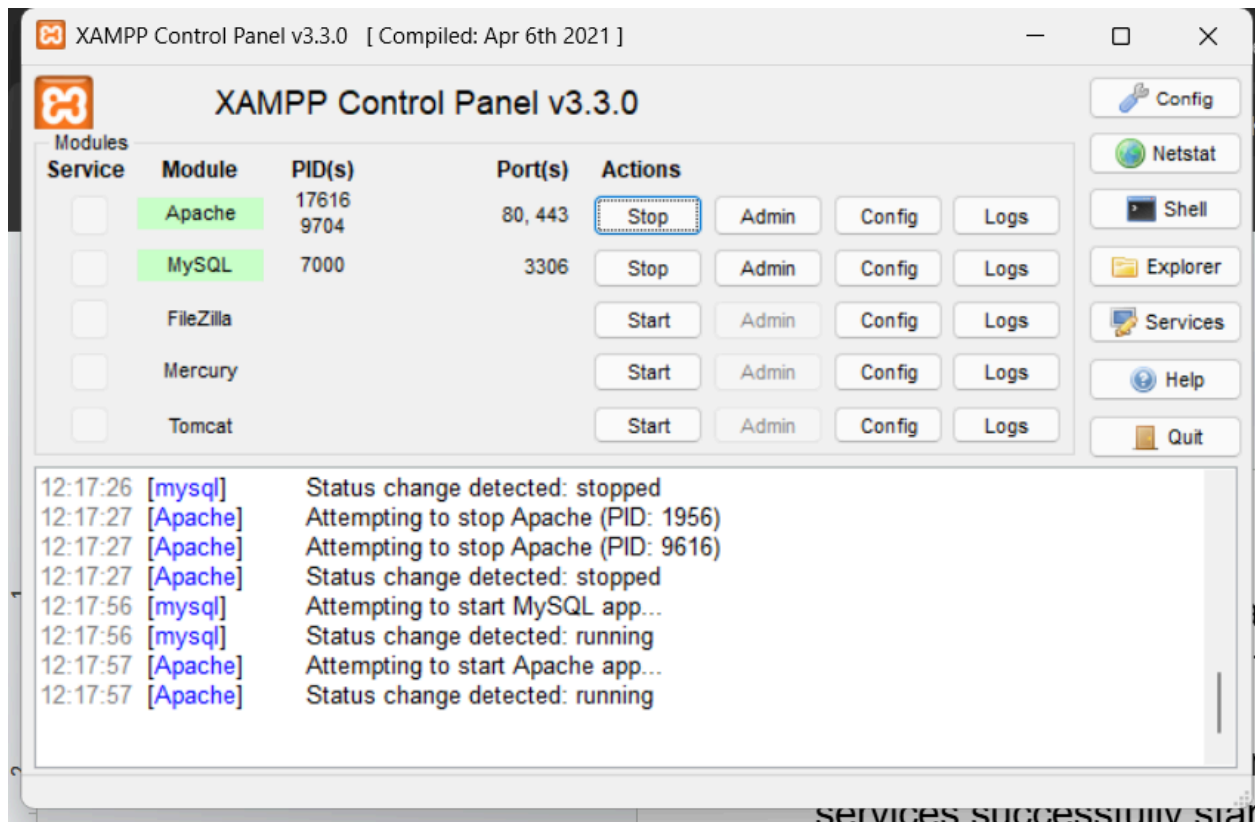
|                                                                                                           |           |
|-----------------------------------------------------------------------------------------------------------|-----------|
| Task 3: CRUD Operations.....                                                                              | 23        |
| Fig 1: Add Records (INSERT).....                                                                          | 23        |
| Fig 2: View Records (SELECT).....                                                                         | 23        |
| Fig 3: Delete Records (DELETE).....                                                                       | 23        |
| Task 4: Connecting PHP to Database.....                                                                   | 24        |
| Fig 1: Database Connection.....                                                                           | 24        |
| Task 5: Fetching Database Records.....                                                                    | 24        |
| Fig 1: Database Records.....                                                                              | 24        |
| <b>Creativity: Note Counter.....</b>                                                                      | <b>25</b> |
| Fig 1: shows the dynamic note count badge in the sidebar header displaying the total number of notes..... | 25        |
| <b>Week 6: Database Integration and CRUD Operations.....</b>                                              | <b>26</b> |
| Practical Task 1.....                                                                                     | 26        |
| Fig 1: Add a Note (INSERT).....                                                                           | 26        |
| Fig 2: View Notes (SELECT).....                                                                           | 26        |
| Fig 3: Edit a Note (UPDATE).....                                                                          | 27        |
| Fig 4: Delete a Note (DELETE).....                                                                        | 27        |
| Fig 5: Database Export.....                                                                               | 27        |
| Practical Task 2.....                                                                                     | 28        |
| Fig 1: Note with Category Added.....                                                                      | 28        |
| Fig 2: Records Stored with Category.....                                                                  | 28        |
| Fig 3: View Records.....                                                                                  | 28        |
| Fig 4: Edit a Record.....                                                                                 | 29        |
| Fig 5: Delete a Record.....                                                                               | 29        |
| Practical Task 3 (Challenge Task).....                                                                    | 30        |
| Fig 1: User Registration.....                                                                             | 30        |
| Fig 2: User Login.....                                                                                    | 30        |
| Fig 3: View Records.....                                                                                  | 31        |
| Fig 4: Search Functionality.....                                                                          | 31        |
| Fig 5: Edit Record.....                                                                                   | 32        |
| Fig 6: Delete Record.....                                                                                 | 32        |
| Fig 7: Responsive Design.....                                                                             | 32        |
| Fig 8: Form Validation.....                                                                               | 33        |

## Week 1: Local Environment Setup

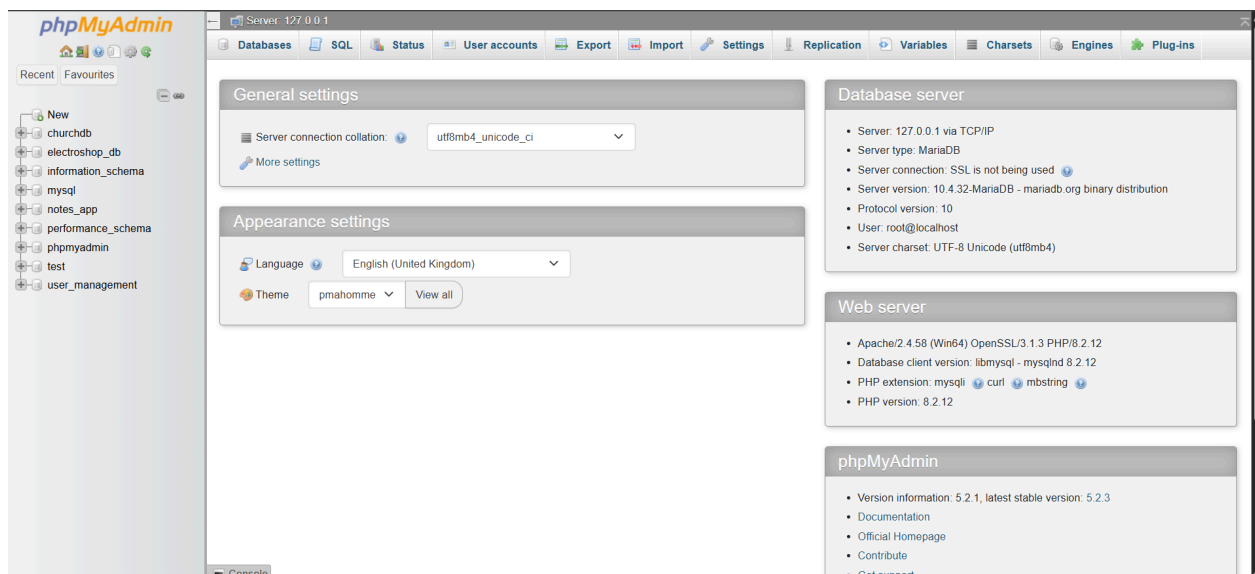
*Fig 1: XAMPP Control Panel showing the local development environment installed on the machine*



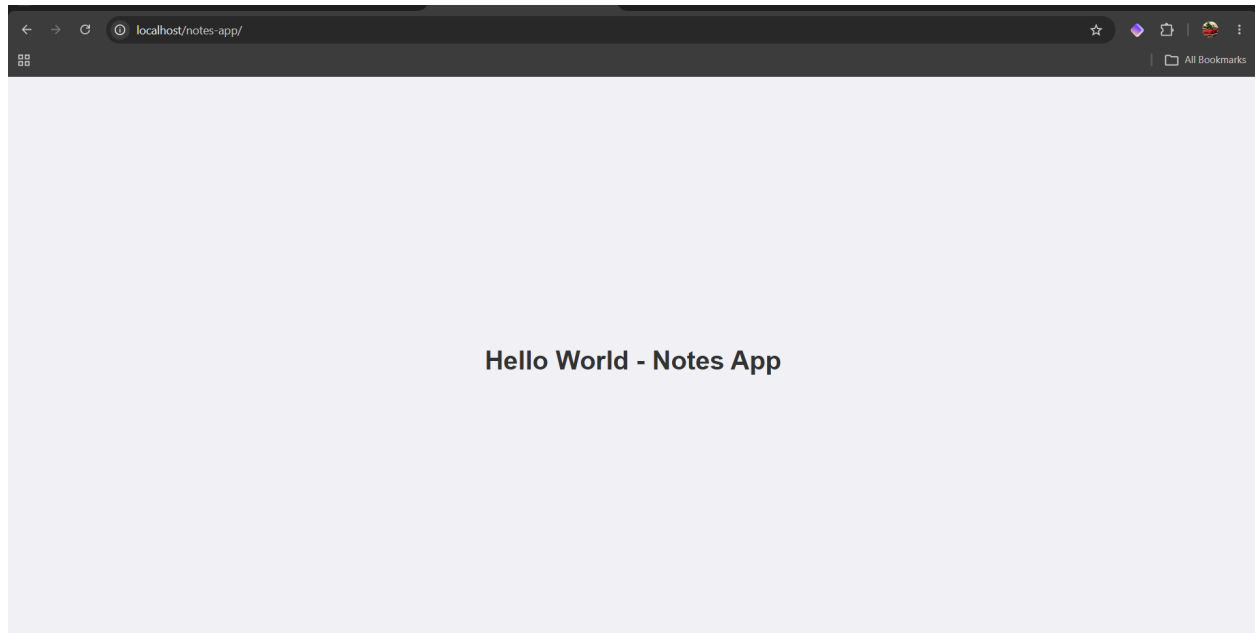
**Fig 2: XAMPP Control Panel showing Apache and MySQL running on their default ports**



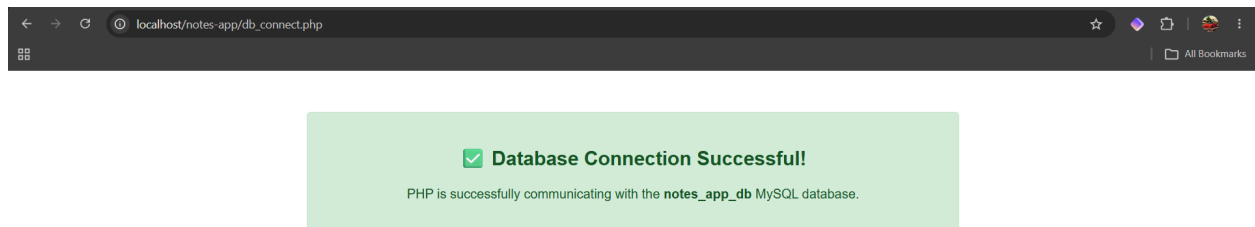
**Fig 3: Browser showing the XAMPP localhost welcome page**



***Fig 4: Hello World Test***

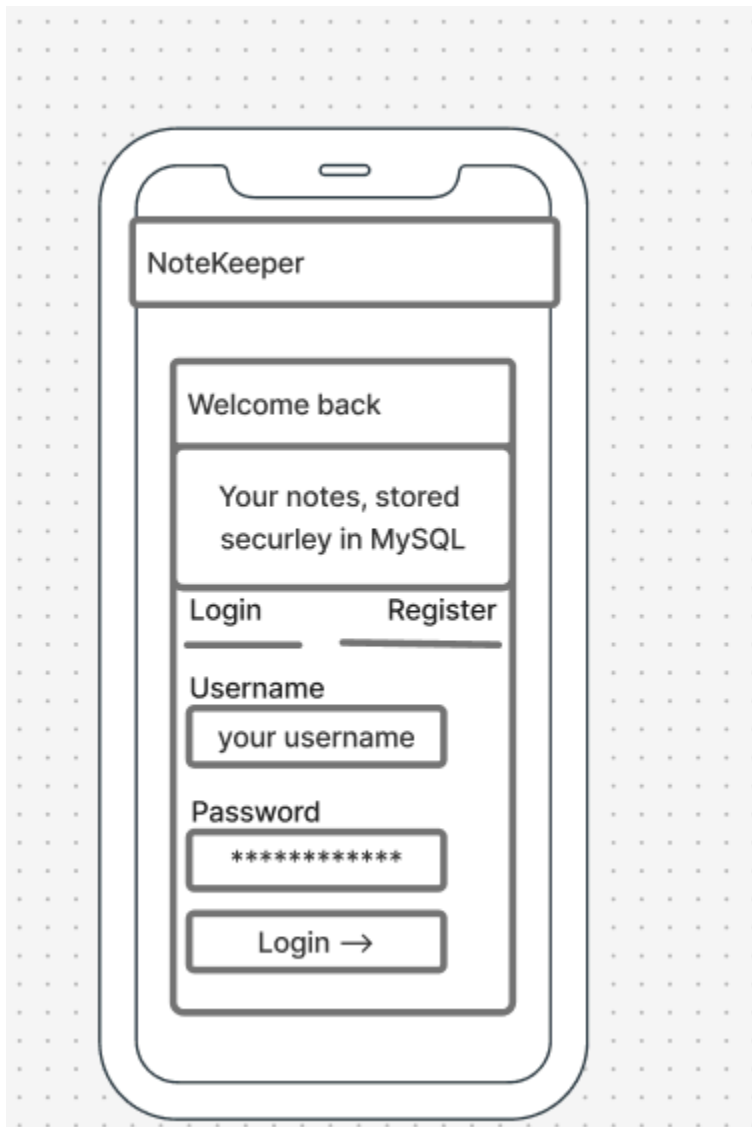


***Fig 5: Database Connection Test***

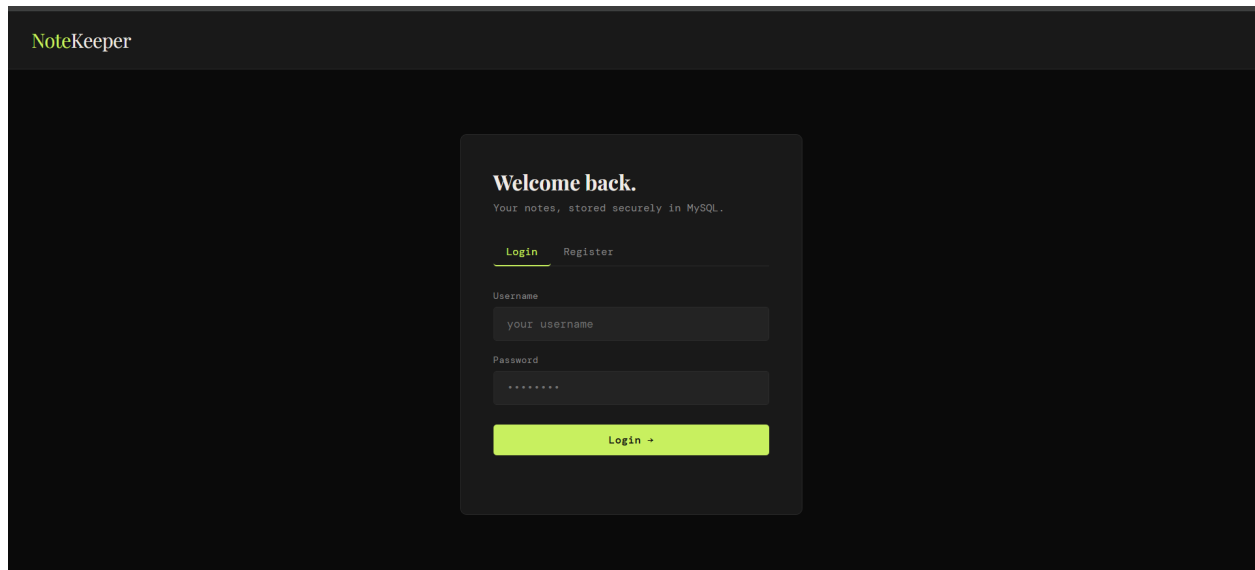


## Week 2: Wireframes and GUI Design

*Fig 1: Figma wireframe detailing the planned layout and user interface for the NoteKeeper login page.*



**Fig 2: Browser showing the Login page with username and password fields used to authenticate registered users**





**Fig 3: Mobile View Mockup**

MY NOTES 2

**Read Hacker News**  
Go to the latest article of The Hacker News  
5/28/2026

**TryHackMe TODAY**  
Do labs on TryHackMe to better understanding  
5/21/2026

**New Note**

**CREATE NOTE**

Title

Note title...

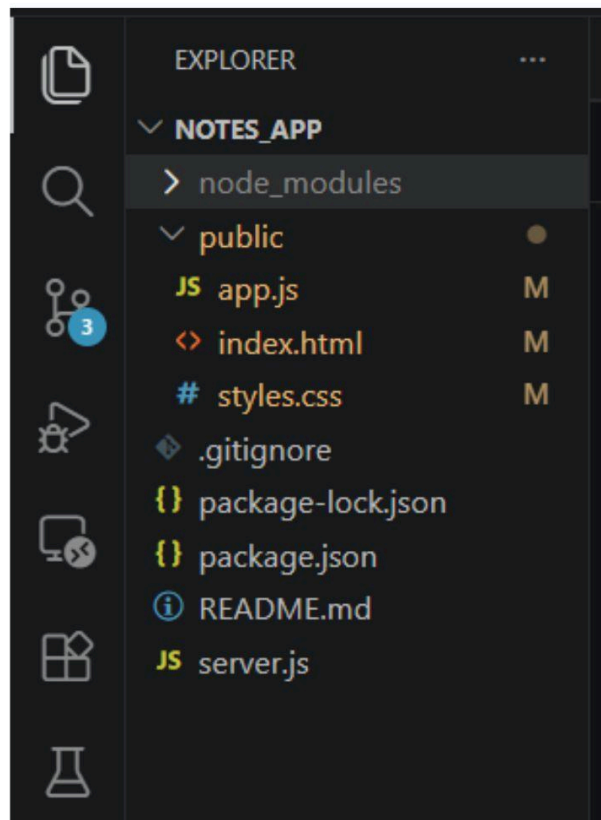
Content

Write your note here...

Save Note

**Fig 4: Color Theme and Navigation Structure**

|                                                                                   |            |
|-----------------------------------------------------------------------------------|------------|
|  | Background |
|  | Surface    |
|  | Accent     |
|  | Text       |
|  | Muted      |
|  | Error      |



## Week 3: JavaScript and PHP Basics

**Fig1: Javascript Form Validation**

**Welcome back.**

Your notes, stored securely in MySQL.

Login **Register**

---

Username

Email

Password

**Create Account →**

Please fill in all fields

**Fig 2: Password Strength Checker**

The image shows a dark-themed web interface for a login and registration system. At the top, it says "Welcome back." followed by "Your notes, stored securely in MySQL." Below this are two tabs: "Login" and "Register", with "Register" being the active tab. The form contains three input fields: "Username" with the value "linet", "Email" with the value "linetkamathi@gmail.com", and "Password" which is masked with dots. Below the password field is a strength indicator consisting of a horizontal bar that is approximately one-third filled with orange, and the word "Medium" below it. At the bottom is a large orange button labeled "Create Account →".

**Welcome back.**

Your notes, stored securely in MySQL.

Login Register

Username

linet

Email

linetkamathi@gmail.com

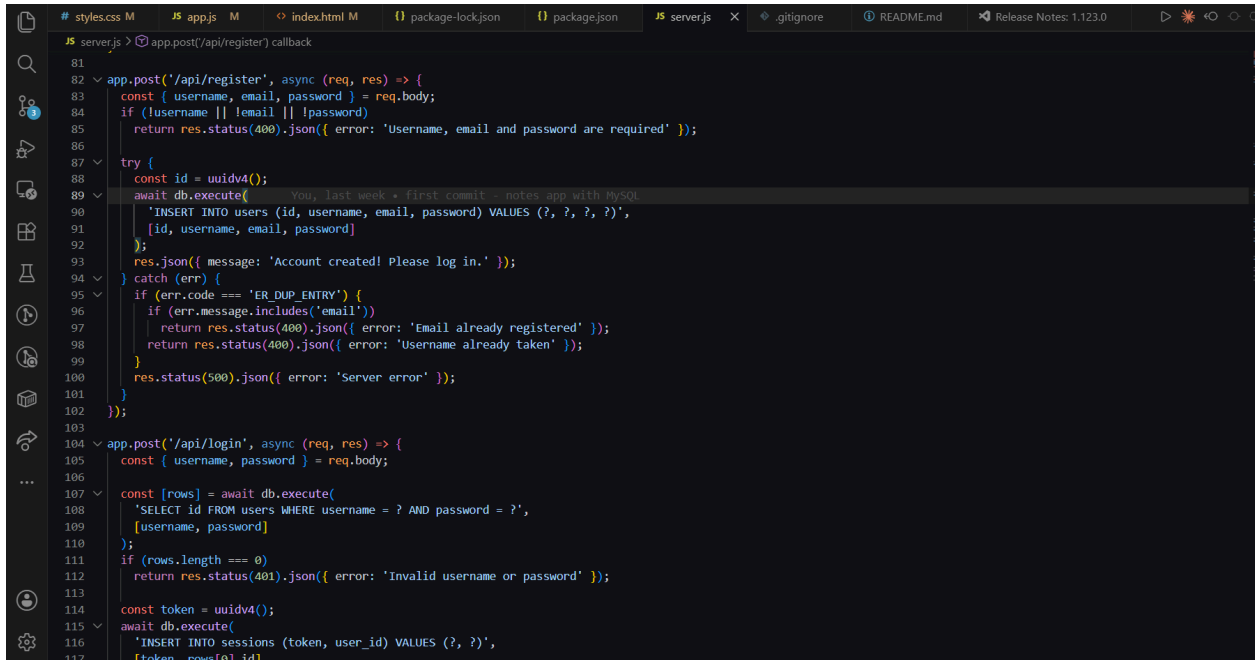
Password

.....

Medium

Create Account →

**Fig 3: Shows the Express.js route syntax used in server.js to handle HTTP requests**

A screenshot of a code editor with a dark theme. The editor shows a file named 'server.js' with JavaScript code for an Express.js application. The code includes two main routes: a registration route at '/api/register' and a login route at '/api/login'. The registration route uses 'app.post' and includes logic to validate user input, create a new user in the database, and return a success message. The login route also uses 'app.post' and includes logic to verify user credentials and return a token. The code is well-formatted with syntax highlighting and line numbers. The editor's interface includes a sidebar with icons for file explorer, search, and other development tools. The top of the editor shows several open files, including 'styles.css', 'app.js', 'index.html', 'package-lock.json', 'package.json', 'server.js', '.gitignore', 'README.md', and 'Release Notes: 1.123.0'.

```
1 # styles.css M JS app.js M index.html M package-lock.json package.json JS server.js X .gitignore README.md Release Notes: 1.123.0
2 JS server.js > app.post('/api/register') callback
3
4 81
5 82 app.post('/api/register', async (req, res) => {
6 83   const { username, email, password } = req.body;
7 84   if (!username || !email || !password)
8 85     return res.status(400).json({ error: 'Username, email and password are required' });
9 86
10 87   try {
11 88     const id = uuidv4();
12 89     await db.execute(
13 90       'INSERT INTO users (id, username, email, password) VALUES (?, ?, ?, ?)',
14 91       [id, username, email, password]
15 92     );
16 93     res.json({ message: 'Account created! Please log in.' });
17 94   } catch (err) {
18 95     if (err.code === 'ER_DUP_ENTRY') {
19 96       if (err.message.includes('email'))
20 97         return res.status(400).json({ error: 'Email already registered' });
21 98       return res.status(400).json({ error: 'Username already taken' });
22 99     }
23 100     res.status(500).json({ error: 'Server error' });
24 101   }
25 102 });
26 103
27 104 app.post('/api/login', async (req, res) => {
28 105   const { username, password } = req.body;
29 106
30 107   const [rows] = await db.execute(
31 108     'SELECT id FROM users WHERE username = ? AND password = ?',
32 109     [username, password]
33 110   );
34 111   if (rows.length === 0)
35 112     return res.status(401).json({ error: 'Invalid username or password' });
36 113
37 114   const token = uuidv4();
38 115   await db.execute(
39 116     'INSERT INTO sessions (token, user_id) VALUES (?, ?)',
40 117     [token, rows[0].id]
```

**Fig 4 shows the database connection configuration and initDB function in server.js.**

```
const DB_CONFIG = {
  host: 'localhost',
  port: 3306,
  user: 'root',
  password: '',
  database: 'notes_app',
  multipleStatements: true
};

let db;

async function initDB() {
  const tempConn = await mysql.createConnection({
    host: DB_CONFIG.host,
    port: DB_CONFIG.port,
    user: DB_CONFIG.user,
    password: DB_CONFIG.password
  });

  await tempConn.execute(`CREATE DATABASE IF NOT EXISTS \`${DB_CONFIG.database}\``);
  await tempConn.end();

  db = await mysql.createPool(DB_CONFIG);
}
```

**Fig 5: Dynamic User Input Handling**

Showing rows 0 - 2 (3 total, Query took 0.0007 seconds)

SELECT \* FROM `notes`

Profiling

Edit inline

Edit

Explain SQL

Create PHP code

Refresh

Show all

Number of rows: 25

Filter rows: Search this table

Sort by key: None

Extra options

|                          |                                                            | id                                   | user_id                              | title            | content                                      | created_at          |
|--------------------------|------------------------------------------------------------|--------------------------------------|--------------------------------------|------------------|----------------------------------------------|---------------------|
| <input type="checkbox"/> | <div><div>Edit</div><div>Copy</div><div>Delete</div></div> | 14f49b21-31f0-4204-aafd-8f7d2e02a11a | 635e5f65-f7fe-4e86-b56d-9a9086dfcf09 | Machine Learning | How Kenya has integrated Machine Learning    | 2026-06-03 12:37:30 |
| <input type="checkbox"/> | <div><div>Edit</div><div>Copy</div><div>Delete</div></div> | 9d568126-c31e-45a2-b458-8d103f3f9c43 | 635e5f65-f7fe-4e86-b56d-9a9086dfcf09 | TryHackMe        | Do labs on TryHackMe to better understanding | 2026-05-21 13:25:24 |
| <input type="checkbox"/> | <div><div>Edit</div><div>Copy</div><div>Delete</div></div> | 9f938a0c-f96-4a76-bdc0-674a8319eb31  | 635e5f65-f7fe-4e86-b56d-9a9086dfcf09 | Read Hacker News | Go to the latest article of The Hacker News  | 2026-05-28 14:33:14 |

# Week 4 Task Schedule

## Task 1 - Understanding Server-Side Programming

Fig 1: Simple Processing Page

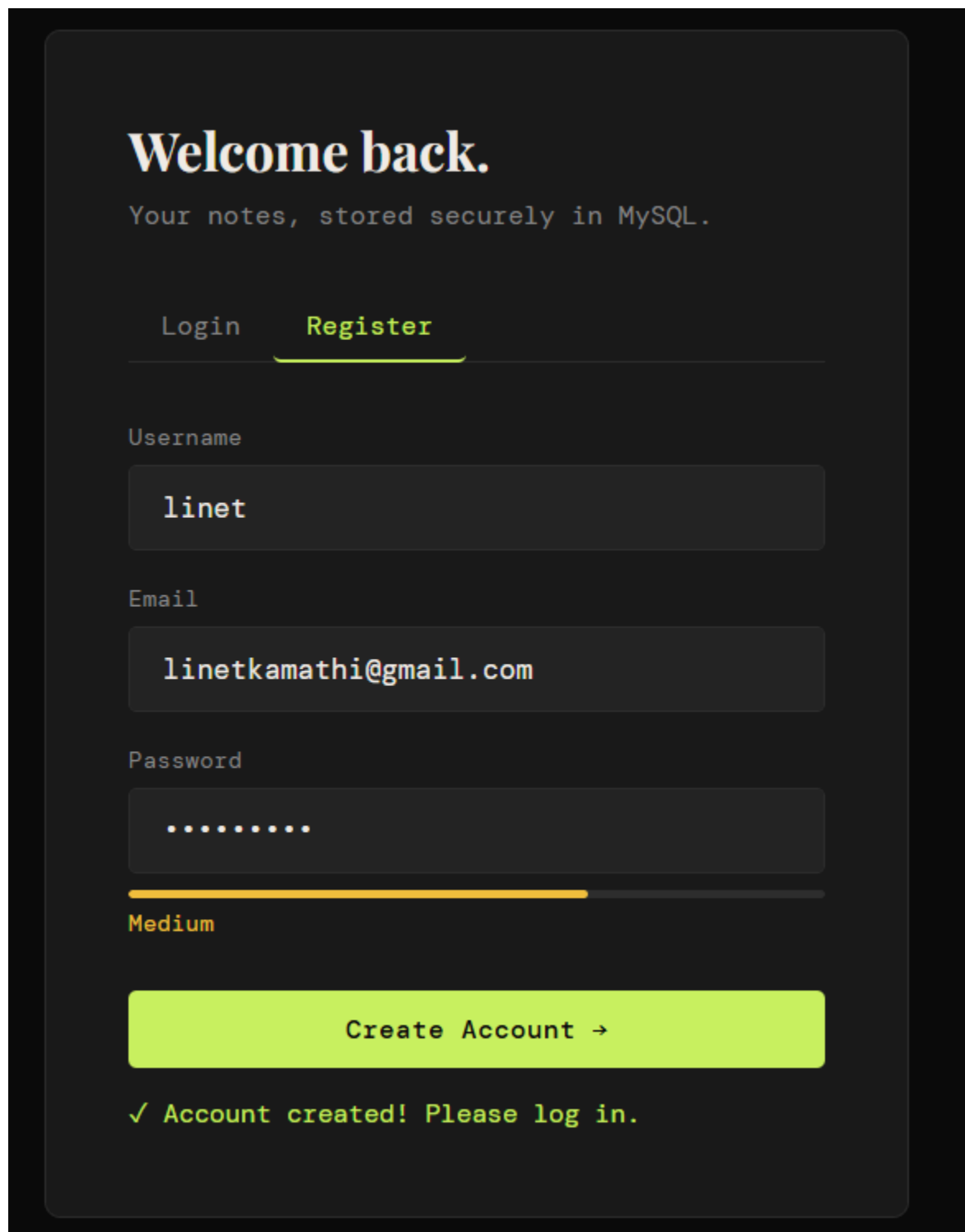
```
app.post('/api/register', async (req, res) => {
  const { username, email, password } = req.body;
  if (!username || !email || !password)
    return res.status(400).json({ error: 'Username, email and password are required' });

  try {
    const id = uuidv4();
    await db.execute(
      'INSERT INTO users (id, username, email, password) VALUES (?, ?, ?, ?)',
      [id, username, email, password]
    );
    res.json({ message: 'Account created! Please log in.' });
  } catch (err) {
    if (err.code === 'ER_DUP_ENTRY') {
      if (err.message.includes('email'))
        return res.status(400).json({ error: 'Email already registered' });
      return res.status(400).json({ error: 'Username already taken' });
    }
    res.status(500).json({ error: 'Server error' });
  }
});
```

Fig 2: Dynamic Welcome Page

The screenshot shows the NoteKeeper web application. At the top, the title 'NoteKeeper' is on the left, and 'logged in as chendler' with a 'logout' button is on the right. A sidebar on the left titled 'MY NOTES' shows a count of 3 and lists three items: 'Machine Learning' (dated 07/08/2020), 'Read Hacker News' (dated 07/08/2020), and 'TryHackMe' (dated 07/08/2020). The main content area is titled 'New Note' and contains a 'CREATE NOTE' form. The form has two input fields: 'Title' with placeholder text 'Note title...' and 'Content' with placeholder text 'Write your note here...'. A green 'Save Note' button is at the bottom of the form.

Fig3: Form Submission System



**Welcome back.**

Your notes, stored securely in MySQL.

Login **Register**

Username

linet

Email

linetkamathi@gmail.com

Password

.....

Medium

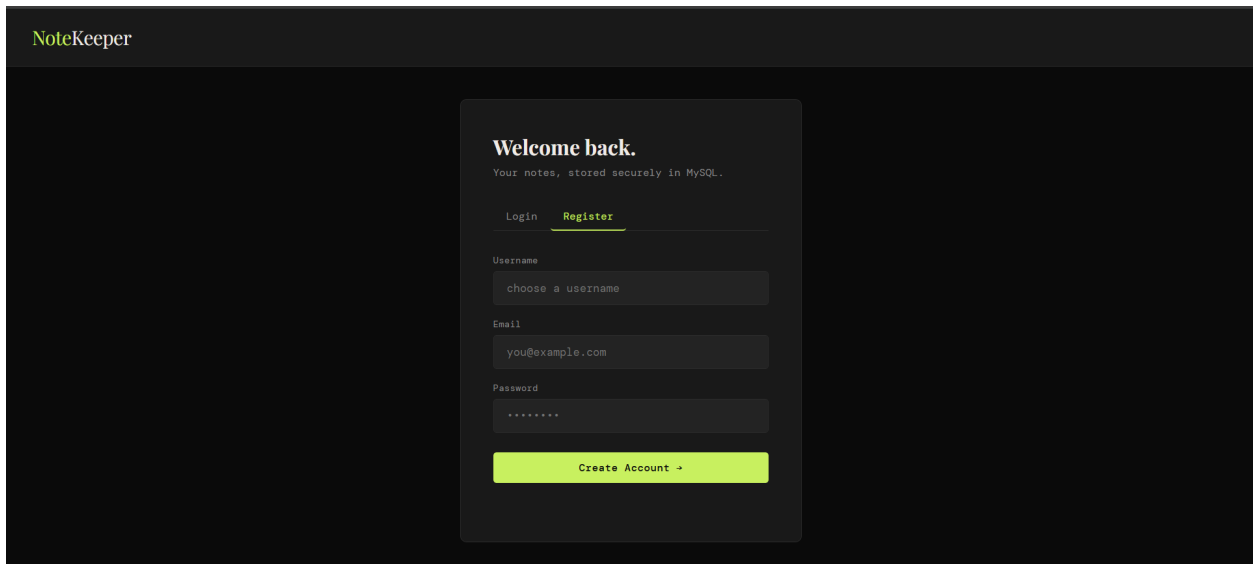
Create Account →

✓ Account created! Please log in.



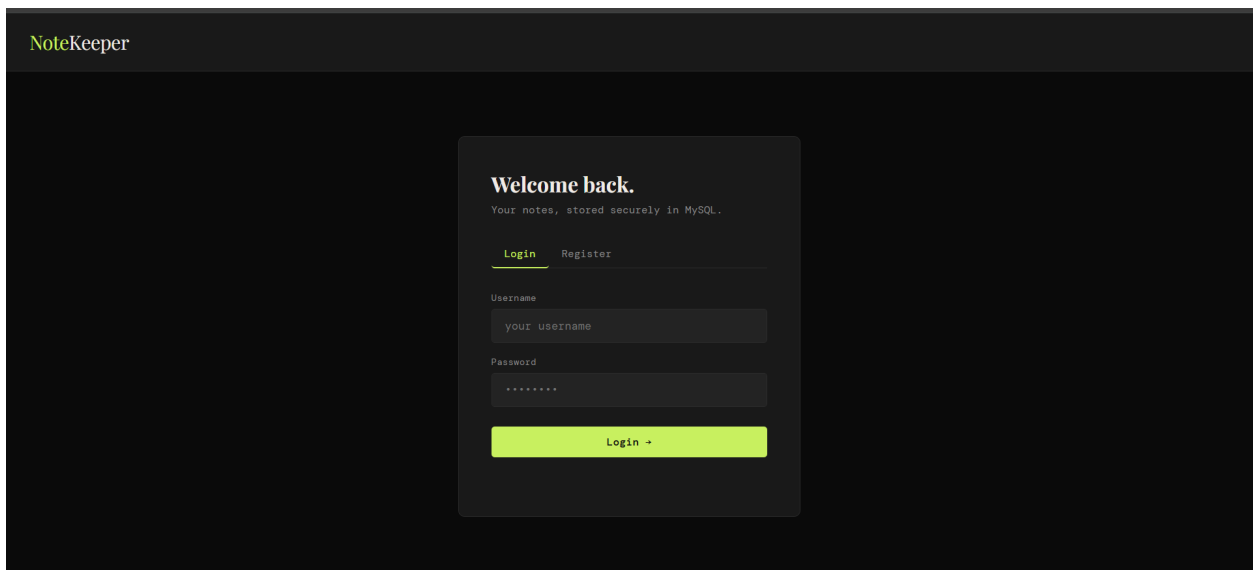
## Task 2 - HTML Forms and PHP Integration

Fig 1: Registration Form



The screenshot shows the registration form of the NoteKeeper application. The header is dark gray with the 'NoteKeeper' logo in green and white. The main content area is black, featuring a central white card. The card has a heading 'Welcome back.' and a subtext 'Your notes, stored securely in MySQL.'. Below this, there are two tabs: 'Login' and 'Register', with 'Register' being the active tab. The form fields are labeled 'Username', 'Email', and 'Password'. The 'Username' field contains the placeholder text 'choose a username'. The 'Email' field contains the placeholder text 'you@example.com'. The 'Password' field contains a series of dots. At the bottom of the card, there is a green button labeled 'Create Account →'.

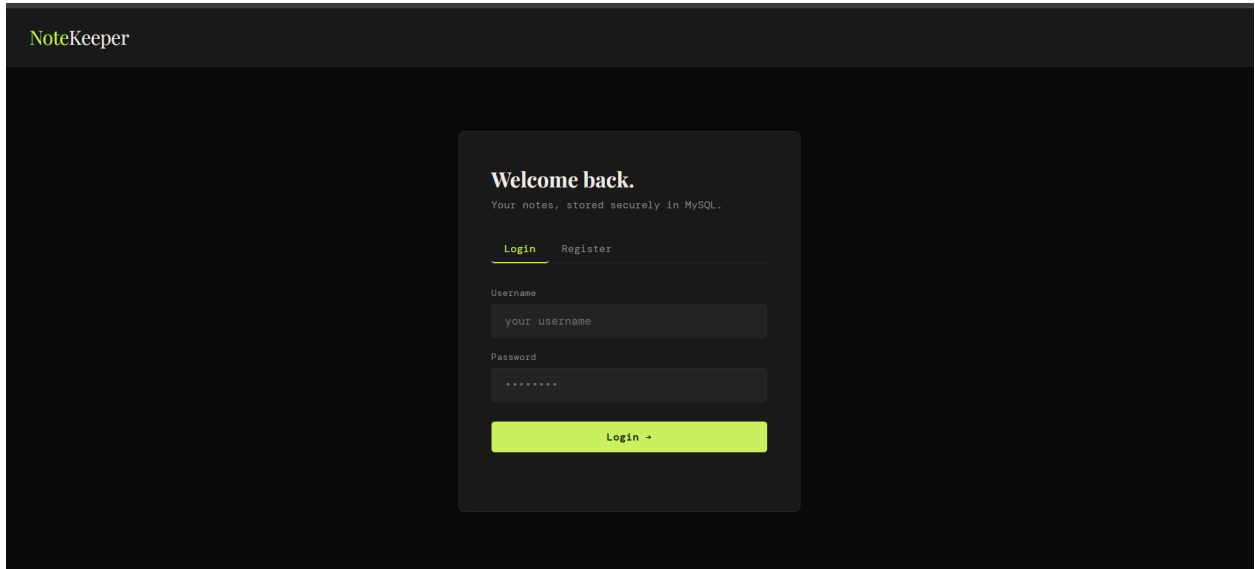
Fig 2: Login Form



The screenshot shows the login form of the NoteKeeper application. The header is dark gray with the 'NoteKeeper' logo in green and white. The main content area is black, featuring a central white card. The card has a heading 'Welcome back.' and a subtext 'Your notes, stored securely in MySQL.'. Below this, there are two tabs: 'Login' and 'Register', with 'Login' being the active tab. The form fields are labeled 'Username' and 'Password'. The 'Username' field contains the placeholder text 'your username'. The 'Password' field contains a series of dots. At the bottom of the card, there is a green button labeled 'Login →'.

## Task 3 - Simple Authentication

Fig 1: Basic Login Page



The image shows a web application interface for 'NoteKeeper'. The header is dark grey with the 'NoteKeeper' logo in the top left corner. The main content area is black and features a central white login card. The card has a heading 'Welcome back.' followed by the text 'Your notes, stored securely in MySQL.'. Below this, there are two links: 'Login' (underlined) and 'Register'. The 'Login' section includes a 'Username' label, a text input field containing 'your username', a 'Password' label, a password input field with masked characters, and a bright green 'Login →' button.

NoteKeeper

**Welcome back.**  
Your notes, stored securely in MySQL.

[Login](#) [Register](#)

Username  
your username

Password  
\*\*\*\*\*

Login →

**Fig 2: Username/passwords validation**

**Welcome back.**

Your notes, stored securely in MySQL.

[Login](#) [Register](#)

Username

linet

Email

linetkamathi@gmail.com

Password

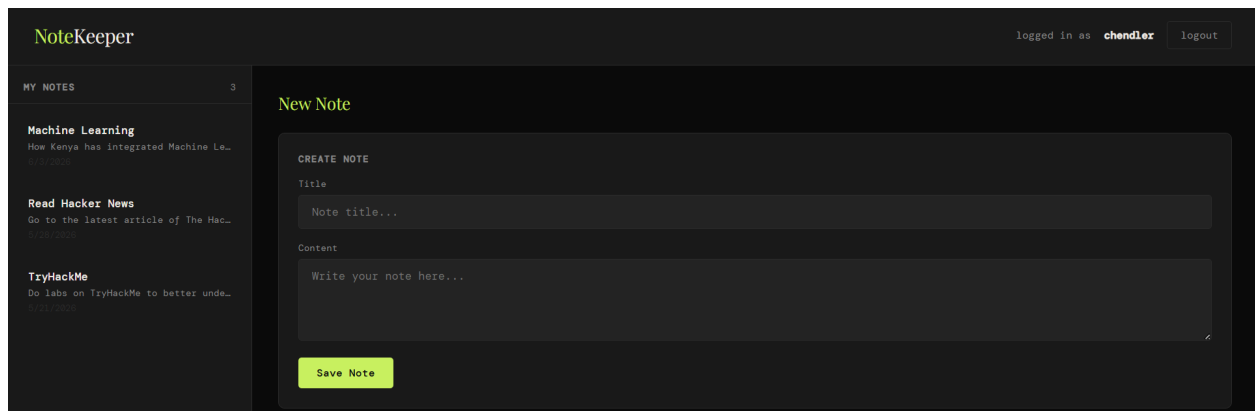
.....

Medium

Create Account →

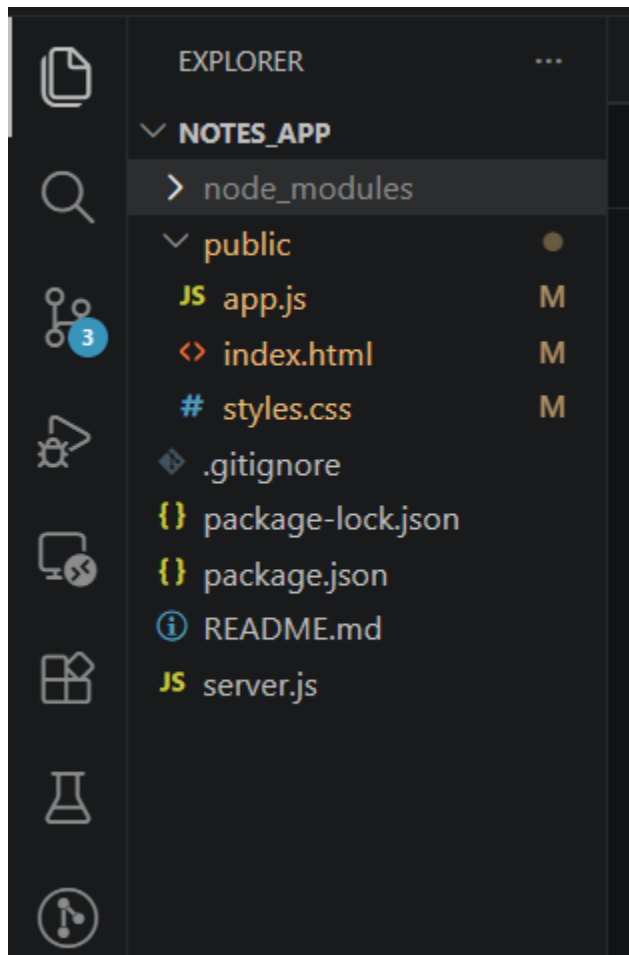
✓ Account created! Please log in.

**Fig 3: Session-based welcome page**



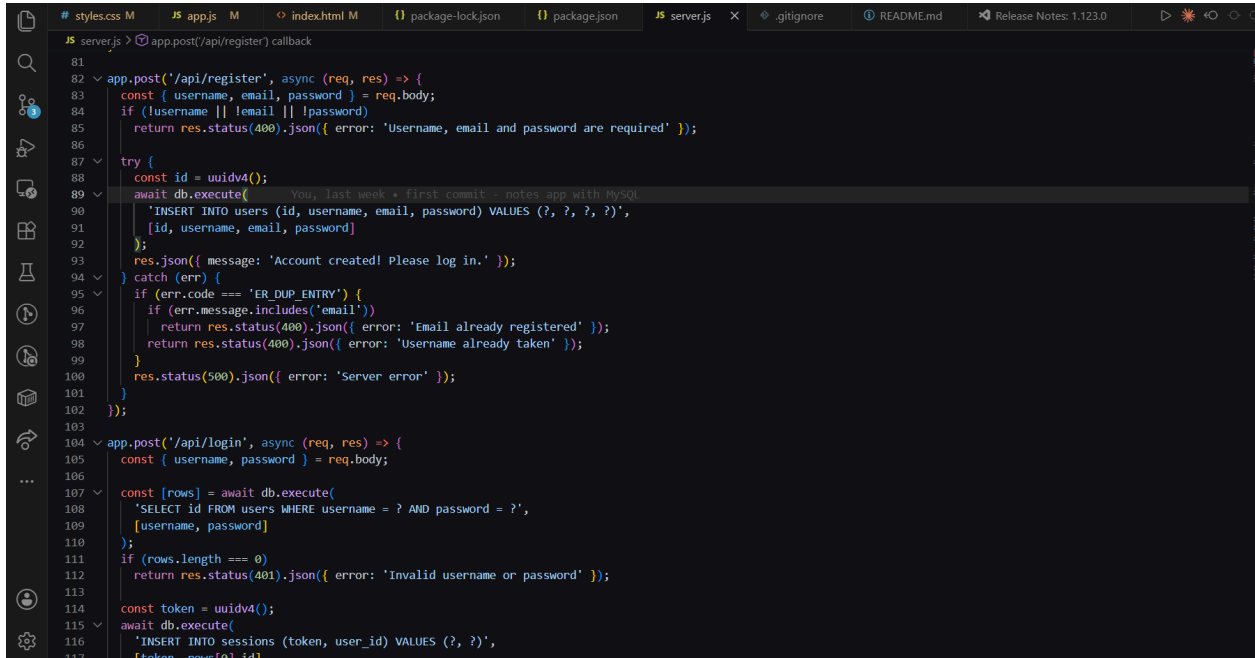
## Task 4 - Backend Folder Organization

**Fig 1: Project Structure**



## Task 5 - Introduction to Modern Backend Alternatives

Fig : PHP Modern Alternative ([Node.js](#))



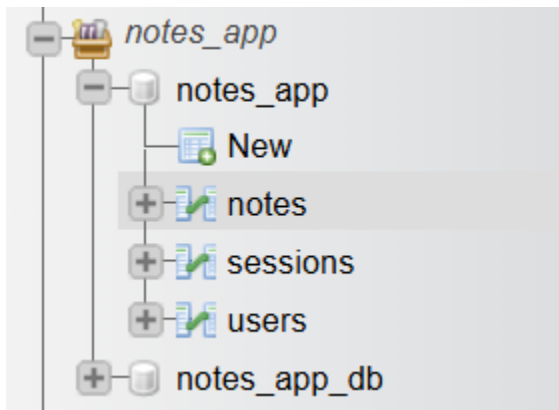
The screenshot shows a code editor with a dark theme. The top bar displays several open files: # styles.css M, JS app.js M, index.html M, package-lock.json, package.json, JS server.js, .gitignore, README.md, and Release Notes: 1.123.0. The main editor area shows the content of server.js, which is a Node.js application using Express.js and MySQL. The code includes a registration endpoint (app.post('/api/register')) and a login endpoint (app.post('/api/login')). The registration endpoint uses uuidv4() for generating unique IDs and db.execute() for inserting user data into a MySQL database. It also includes error handling for duplicate entries, email already registered, and username already taken. The login endpoint uses db.execute() to query the users table and db.execute() to insert a session token into a sessions table. The code is written in JavaScript with ES6 syntax, including arrow functions and const/let for variable declarations.

```
81
82 app.post('/api/register', async (req, res) => {
83   const { username, email, password } = req.body;
84   if (!username || !email || !password)
85     return res.status(400).json({ error: 'Username, email and password are required' });
86
87   try {
88     const id = uuidv4();
89     await db.execute(`
90       INSERT INTO users (id, username, email, password) VALUES (?, ?, ?, ?),
91       [id, username, email, password]
92     `);
93     res.json({ message: 'Account created! Please log in.' });
94   } catch (err) {
95     if (err.code === 'ER_DUP_ENTRY') {
96       if (err.message.includes('email'))
97         return res.status(400).json({ error: 'Email already registered' });
98       return res.status(400).json({ error: 'Username already taken' });
99     }
100    res.status(500).json({ error: 'Server error' });
101  }
102 });
103
104 app.post('/api/login', async (req, res) => {
105   const { username, password } = req.body;
106
107   const [rows] = await db.execute(
108     'SELECT id FROM users WHERE username = ? AND password = ?',
109     [username, password]
110   );
111   if (rows.length === 0)
112     return res.status(401).json({ error: 'Invalid username or password' });
113
114   const token = uuidv4();
115   await db.execute(
116     'INSERT INTO sessions (token, user_id) VALUES (?, ?)',
117     [token, rows[0].id]
```

# Week 5 - Advanced Web Design and Development

## Task 1: Database Creation

Fig 1: notes-app database



## Task 2: Creating Tables

Fig 1: notes-app tables

| Table                             | Action                                      | Rows | Type   | Collation          | Size      | Overhead |
|-----------------------------------|---------------------------------------------|------|--------|--------------------|-----------|----------|
| <input type="checkbox"/> notes    | ★ Browse Structure Search Insert Empty Drop | 3    | InnoDB | utf8mb4_general_ci | 32.0 KiB  | -        |
| <input type="checkbox"/> sessions | ★ Browse Structure Search Insert Empty Drop | 2    | InnoDB | utf8mb4_general_ci | 32.0 KiB  | -        |
| <input type="checkbox"/> users    | ★ Browse Structure Search Insert Empty Drop | 3    | InnoDB | utf8mb4_general_ci | 48.0 KiB  | -        |
| 3 tables                          | Sum                                         | 8    | InnoDB | utf8mb4_general_ci | 112.0 KiB | 0 B      |

# Task 3: CRUD Operations

Fig 1: Add Records (INSERT)

| + -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                        |                                                                                        | id                                                                                       | user_id                              | title                                | content          | created_at                                   |                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------------------------------------|--------------------------------------|------------------|----------------------------------------------|---------------------|
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  Edit |  Copy |  Delete | 14f49b21-31f0-4204-aafd-8f7d2e02a11a | 635e5f65-f7fe-4e86-b56d-9a9086dfcf09 | Machine Learning | How Kenya has integrated Machine Learning    | 2026-06-03 12:37:30 |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  Edit |  Copy |  Delete | 9d568126-c31e-45a2-b458-8d103f3f9c43 | 635e5f65-f7fe-4e86-b56d-9a9086dfcf09 | TryHackMe        | Do labs on TryHackMe to better understanding | 2026-05-21 13:25:24 |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  Edit |  Copy |  Delete | 9f938a0c-ff96-4a76-bdc0-674a8319eb31 | 635e5f65-f7fe-4e86-b56d-9a9086dfcf09 | Read Hacker News | Go to the latest article of The Hacker News  | 2026-05-28 14:33:14 |
| <div> <input type="checkbox"/> Check all    With selected:  Edit     Copy     Delete     Export</div> |                                                                                        |                                                                                        |                                                                                          |                                      |                                      |                  |                                              |                     |

Fig 2: View Records (SELECT)

NoteKeeper

MY NOTES3

Machine Learning

How Kenya has integrated Machine Le...

Read Hacker News

Go to the latest article of The Hac...

5/28/2026

TryHackMe

Do labs on TryHackMe to better unde...

5/21/2026

New Note

CREATE NOTE

Title

Note title...

Content

Write your note here...

Save Note

Fig 3: Delete Records (DELETE)

|                          |                                                                                                                                                                                                                                                                              | id                                   | user_id                              | title            | content                                      | created_at          |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|--------------------------------------|------------------|----------------------------------------------|---------------------|
| <input type="checkbox"/> |  Edit  Copy  Delete | 9d568126-c31e-45a2-b458-8d103f3f9c43 | 635e5f65-f7fe-4e86-b56d-9a9086dfcf09 | TryHackMe        | Do labs on TryHackMe to better understanding | 2026-05-21 13:25:24 |
| <input type="checkbox"/> |  Edit  Copy  Delete | 9f938a0c-ff96-4a76-bdc0-674a8319eb31 | 635e5f65-f7fe-4e86-b56d-9a9086dfcf09 | Read Hacker News | Go to the latest article of The Hacker News  | 2026-05-28 14:33:14 |

Fig 4: Edit Records (UPDATE)

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                              | id                                   | user_id                              | title            | content                                      | created_at          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|--------------------------------------|------------------|----------------------------------------------|---------------------|
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  Edit  Copy  Delete | 9d568126-c31e-45a2-b458-8d103f3f9c43 | 635e5f65-f7fe-4e86-b56d-9a9086dfcf09 | TryHackMe TODAY  | Do labs on TryHackMe to better understanding | 2026-05-21 13:25:24 |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  Edit  Copy  Delete | 9f938a0c-ff96-4a76-bdc0-674a8319eb31 | 635e5f65-f7fe-4e86-b56d-9a9086dfcf09 | Read Hacker News | Go to the latest article of The Hacker News  | 2026-05-28 14:33:14 |
| <div> <input type="checkbox"/> Check all    With selected:  Edit     Copy     Delete     Export</div> |                                                                                                                                                                                                                                                                              |                                      |                                      |                  |                                              |                     |

## Task 4: Connecting PHP to Database

Fig 1: Database Connection

```
const DB_CONFIG = {
  host: 'localhost',
  port: 3306,
  user: 'root',
  password: '',
  database: 'notes_app',
  multipleStatements: true
};

let db;

async function initDB() {
  const tempConn = await mysql.createConnection({
    host: DB_CONFIG.host,
    port: DB_CONFIG.port,
    user: DB_CONFIG.user,
    password: DB_CONFIG.password
  });

  await tempConn.execute(`CREATE DATABASE IF NOT EXISTS \`${DB_CONFIG.database}\``);
  await tempConn.end();

  db = await mysql.createPool(DB_CONFIG);
}
```

## Task 5: Fetching Database Records

Fig 1: Database Records

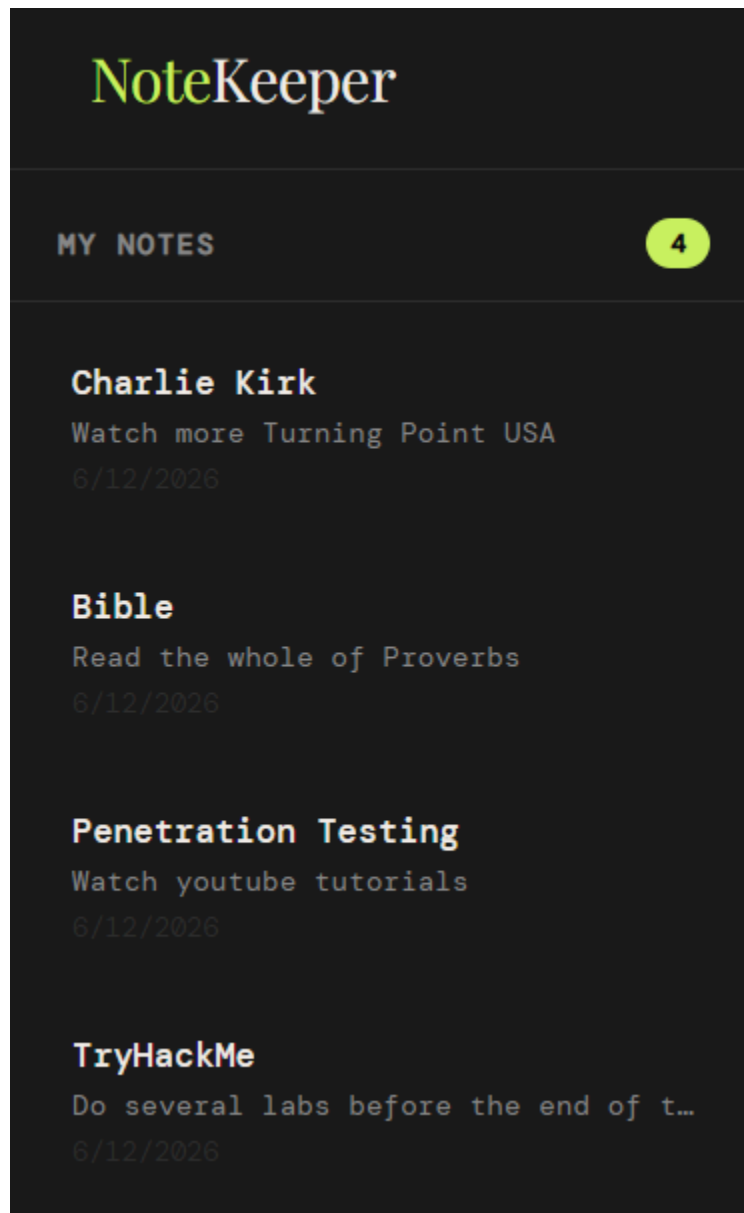
```
app.put('/api/notes/:id', requireAuth, async (req, res) => {
  const { title, content } = req.body;
  if (!title || !content)
    return res.status(400).json({ error: 'Title and content required' });

  await db.execute(
    'UPDATE notes SET title = ?, content = ? WHERE id = ? AND user_id = ?',
    [title, content, req.params.id, req.userId]
  );
  res.json({ message: 'Note updated!' });
});
```



## Creativity: Note Counter

**Fig 1: shows the dynamic note count badge in the sidebar header displaying the total number of notes.**



# Week 6: Database Integration and CRUD Operations

## Practical Task 1

Fig 1: Add a Note (INSERT)

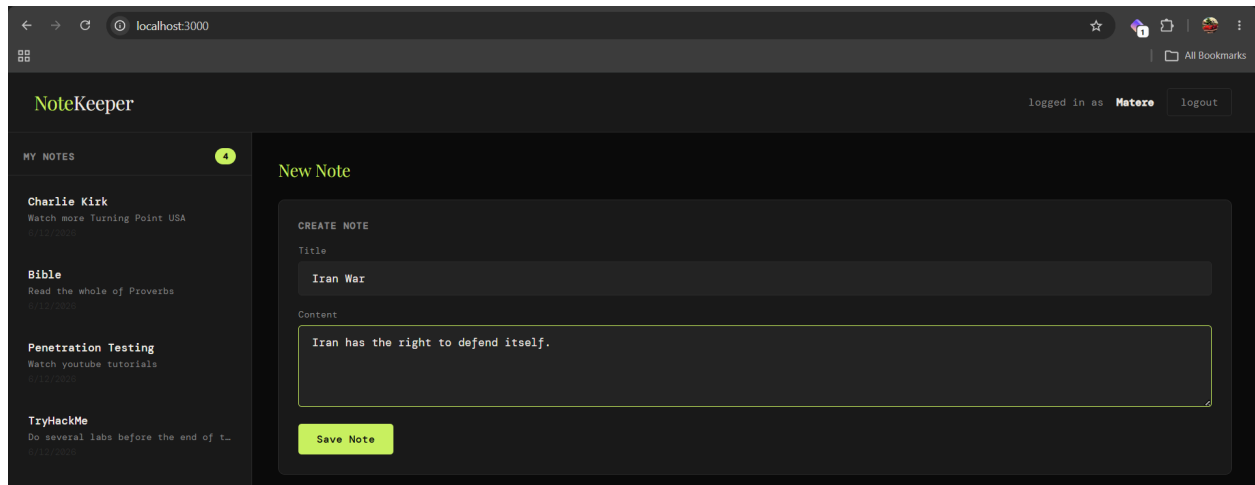
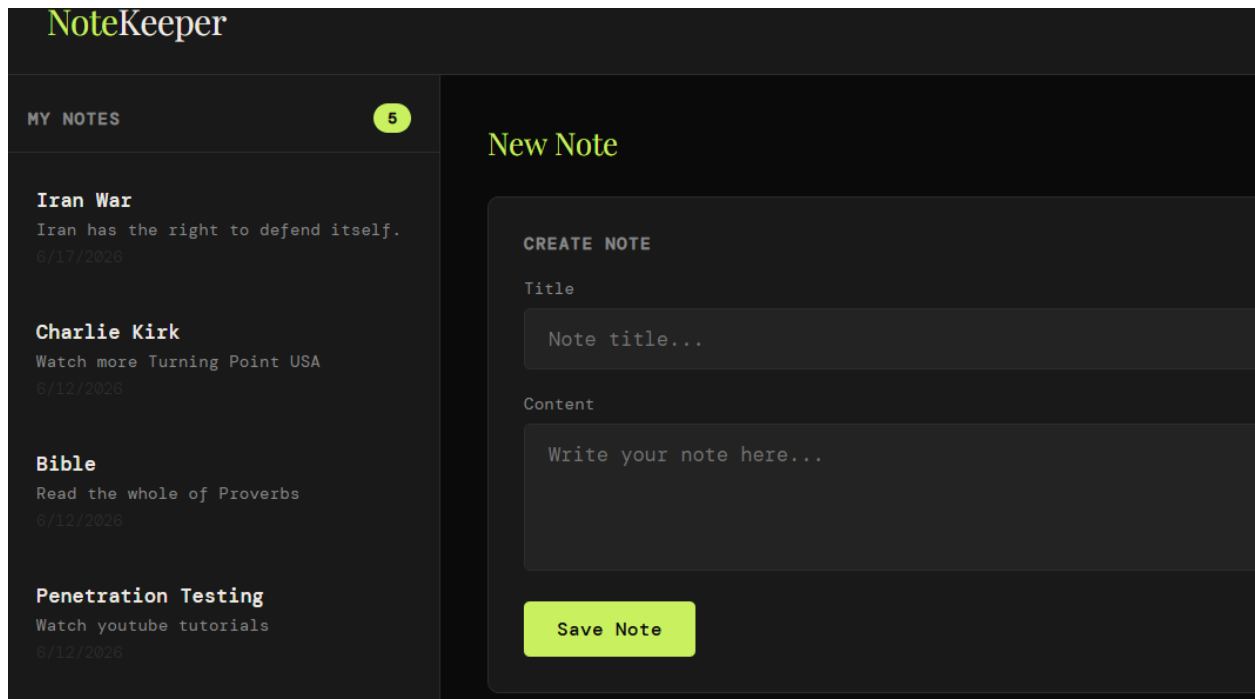


Fig 2: View Notes (SELECT)



**Fig 3: Edit a Note (UPDATE)**

New Note

CREATE NOTE

Title

Pen Tester

Content

Watch youtube tutorials

Update Note

**Fig 4: Delete a Note (DELETE)**

|                          |                    | id                                   | user_id                              | title        | content                                     | created_at          |
|--------------------------|--------------------|--------------------------------------|--------------------------------------|--------------|---------------------------------------------|---------------------|
| <input type="checkbox"/> | Edit  Copy  Delete | 44de630b-b69d-4329-9c6f-ad2a23efb42e | 954c422b-6eb7-486f-92ed-99341205d3f2 | Charlie Kirk | Watch more Turning Point USA                | 2026-06-12 04:53:18 |
| <input type="checkbox"/> | Edit  Copy  Delete | 6daa1bf7-84f3-41fa-af38-648203afca9f | 954c422b-6eb7-486f-92ed-99341205d3f2 | TryHackMe    | Do several labs before the end of the week. | 2026-06-12 04:50:01 |
| <input type="checkbox"/> | Edit  Copy  Delete | 7fe950ea-1a33-4bae-9cce-cf6e33656dca | 954c422b-6eb7-486f-92ed-99341205d3f2 | Iran War     | Iran has the right to defend itself.        | 2026-06-17 07:56:39 |
| <input type="checkbox"/> | Edit  Copy  Delete | bd2baec9-358b-467f-88dd-bd13f521a874 | 954c422b-6eb7-486f-92ed-99341205d3f2 | Bible        | Read the whole of Proverbs                  | 2026-06-12 04:52:54 |

**Fig 5: Database Export**

```
> week1
> week2
> week3
> week4
> week5
JS server.js
.gitignore
Notes Application Docum...
notes_app.sql
package-lock.json

71
72
73
74
75
76
77
78
79
80
81
82

<div class= sidebar
  <div class="sideba
    <h3>My Notes</h3>
    <span id="noteCo
  </div>
  <div class="notes-
    <div class="empt
  </div>
</div>
<div class="main-par
  <div class="panel-
```

## Practical Task 2

Fig 1: Note with Category Added

### New Note

CREATE NOTE

Title

Cipher Text

Category

Study

Content

Why is it called Caesar Cipher.

Save Note

Fig 2: Records Stored with Category

|                                                                                                    |  |      |  | id   | user_id | title  | content                              | category                             | created_at          |                                                   |          |                     |
|----------------------------------------------------------------------------------------------------|--|------|--|------|---------|--------|--------------------------------------|--------------------------------------|---------------------|---------------------------------------------------|----------|---------------------|
| <input type="checkbox"/>                                                                           |  | Edit |  | Copy |         | Delete | 065373d4-ef12-453c-9a9a-43778fa81694 | 954c422b-6eb7-486f-92ed-99341205d3f2 | Bible Study         | Join the Bible study today at 8pm.                | Personal | 2026-06-17 08:29:58 |
| <input type="checkbox"/>                                                                           |  | Edit |  | Copy |         | Delete | 38e99915-e346-424c-857e-4fac777a9a82 | 954c422b-6eb7-486f-92ed-99341205d3f2 | Drink More Water    | Water helps with everyday tasks                   | General  | 2026-06-17 08:28:51 |
| <input type="checkbox"/>                                                                           |  | Edit |  | Copy |         | Delete | 4e52ad21-953c-494e-8829-fae1206837c6 | 954c422b-6eb7-486f-92ed-99341205d3f2 | Tech Companies      | Research about several tech companies in Nairobi. | Work     | 2026-06-17 08:29:36 |
| <input type="checkbox"/>                                                                           |  | Edit |  | Copy |         | Delete | 820f7cd8-5ae5-4865-83ec-692f802c00f3 | 954c422b-6eb7-486f-92ed-99341205d3f2 | Penetration Testing | Watch tutorials on how to Pen Test.               | Study    | 2026-06-17 08:30:30 |
| <div> <input type="checkbox"/> Check all    With selected:  Edit    Copy    Delete    Export</div> |  |      |  |      |         |        |                                      |                                      |                     |                                                   |          |                     |

Fig 3: View Records

MY NOTES

4

Penetration Testing

Watch tutorials on how to Pen Test.

07/17/2026

Bible Study

Join the Bible study today at 8pm.

07/17/2026

Tech Companies

Research about several tech compani...

07/17/2026

Drink More Water

Water helps with everyday tasks.

07/17/2026

### New Note

CREATE NOTE

Title

Note title...

Category

Study

Content

Write your note here...

Save Note

Fig 4: Edit a Record

New Note

CREATE NOTE

Title

Tech Companies & Software

Category

Study

Content

Research about several tech companies in Nairobi.

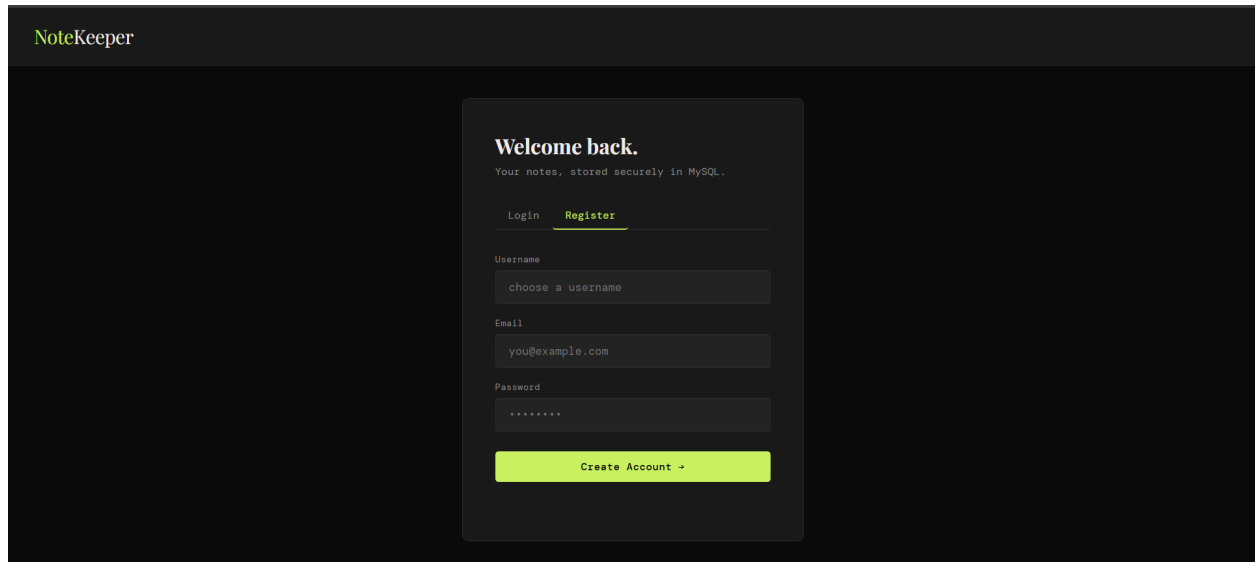
Update Note

Fig 5: Delete a Record

|                                                                                     |                                                                                                                                                                                                                                                                        | id                                                                                                                                                                                                                                                                                                                                                                                                                           | user_id                              | title                     | content                                           | category | created_at          |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|---------------------------|---------------------------------------------------|----------|---------------------|
| <input type="checkbox"/>                                                            |  Edit  Copy  Delete | 065373d4-ef12-453c-9a9a-43778fa81694                                                                                                                                                                                                                                                                                                                                                                                         | 954c422b-6eb7-486f-92ed-99341205d3f2 | Bible Study               | Join the Bible study today at 8pm.                | Personal | 2026-06-17 08:29:58 |
| <input type="checkbox"/>                                                            |  Edit  Copy  Delete | 4e52ad21-953c-494e-8829-fae1206837c6                                                                                                                                                                                                                                                                                                                                                                                         | 954c422b-6eb7-486f-92ed-99341205d3f2 | Tech Companies & Software | Research about several tech companies in Nairobi. | Work     | 2026-06-17 08:29:36 |
| <input type="checkbox"/>                                                            |  Edit  Copy  Delete | 820f7cd8-5ae5-4865-83ec-692f802c00f3                                                                                                                                                                                                                                                                                                                                                                                         | 954c422b-6eb7-486f-92ed-99341205d3f2 | Penetration Testing       | Watch tutorials on how to Pen Test.               | Study    | 2026-06-17 08:30:30 |
|  |                                                                                                                                                                                                                                                                        | <input type="checkbox"/> Check all    With selected:  Edit  Copy  Delete  Export |                                      |                           |                                                   |          |                     |

## Practical Task 3 (Challenge Task)

Fig 1: User Registration



The image shows a user registration form for a web application named "NoteKeeper". The form is centered on a dark background. At the top left, the "NoteKeeper" logo is displayed. The form itself has a title "Welcome back." and a subtitle "Your notes, stored securely in MySQL.". Below this, there are two tabs: "Login" and "Register", with "Register" being the active tab. The form contains three input fields: "Username" with a placeholder "choose a username", "Email" with a placeholder "you@example.com", and "Password" with a placeholder "\*\*\*\*\*". At the bottom of the form is a green button labeled "Create Account →".

NoteKeeper

Welcome back.  
Your notes, stored securely in MySQL.

Login Register

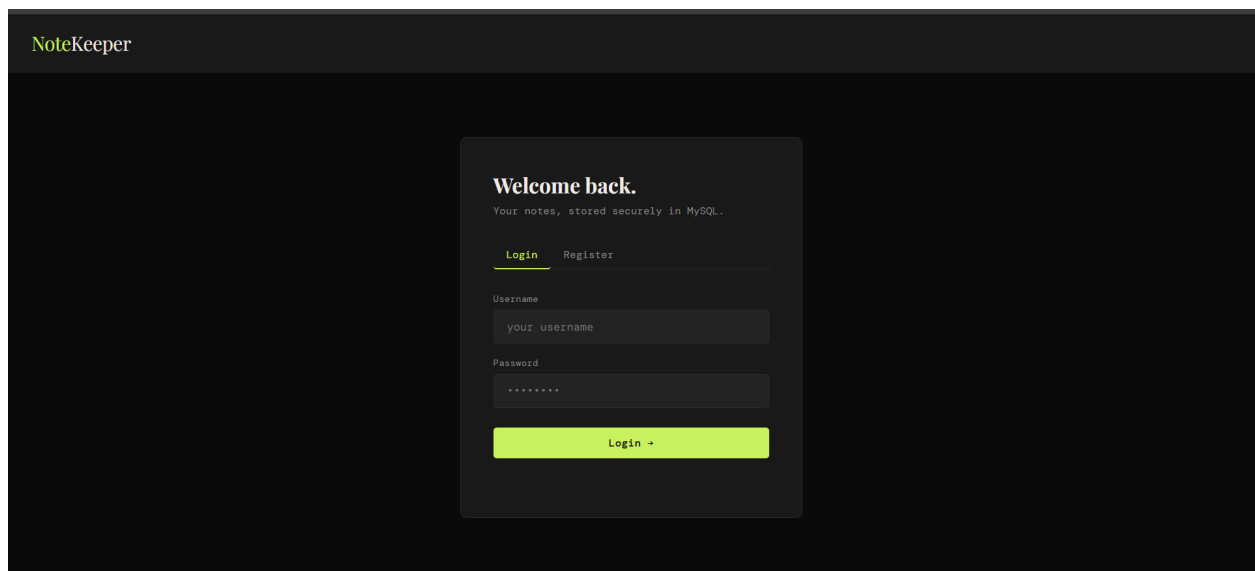
Username  
choose a username

Email  
you@example.com

Password  
\*\*\*\*\*

Create Account →

Fig 2: User Login



The image shows a user login form for the same "NoteKeeper" web application. The form is centered on a dark background. At the top left, the "NoteKeeper" logo is displayed. The form has a title "Welcome back." and a subtitle "Your notes, stored securely in MySQL.". Below this, there are two tabs: "Login" and "Register", with "Login" being the active tab. The form contains two input fields: "Username" with a placeholder "your username" and "Password" with a placeholder "\*\*\*\*\*". At the bottom of the form is a green button labeled "Login →".

NoteKeeper

Welcome back.  
Your notes, stored securely in MySQL.

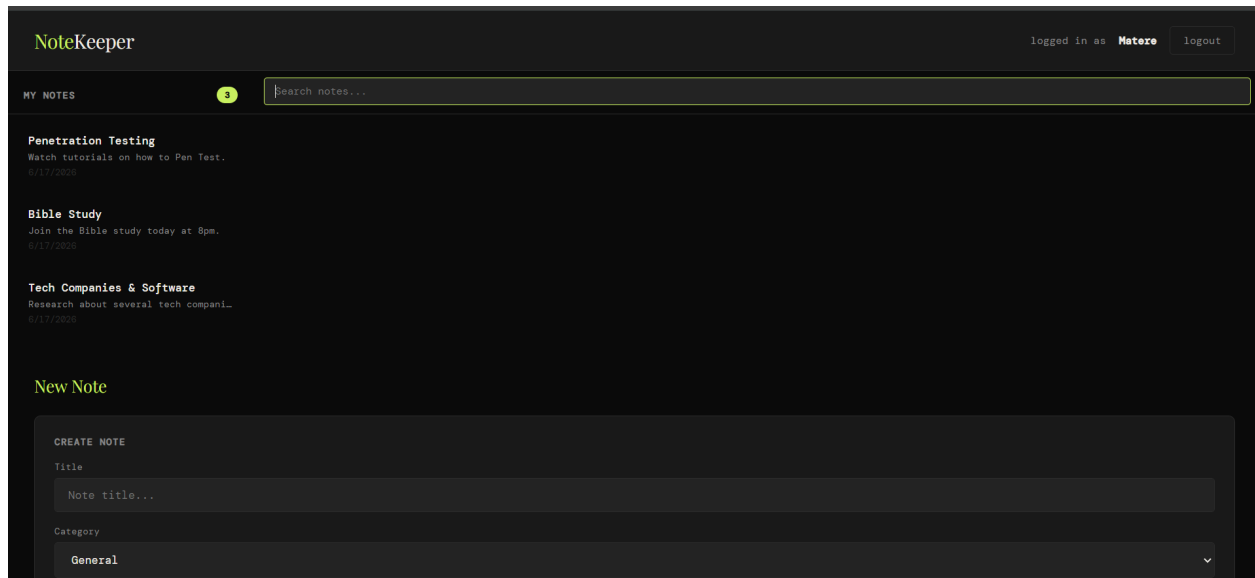
Login Register

Username  
your username

Password  
\*\*\*\*\*

Login →

**Fig 3: View Records**



**Fig 4: Search Functionality**

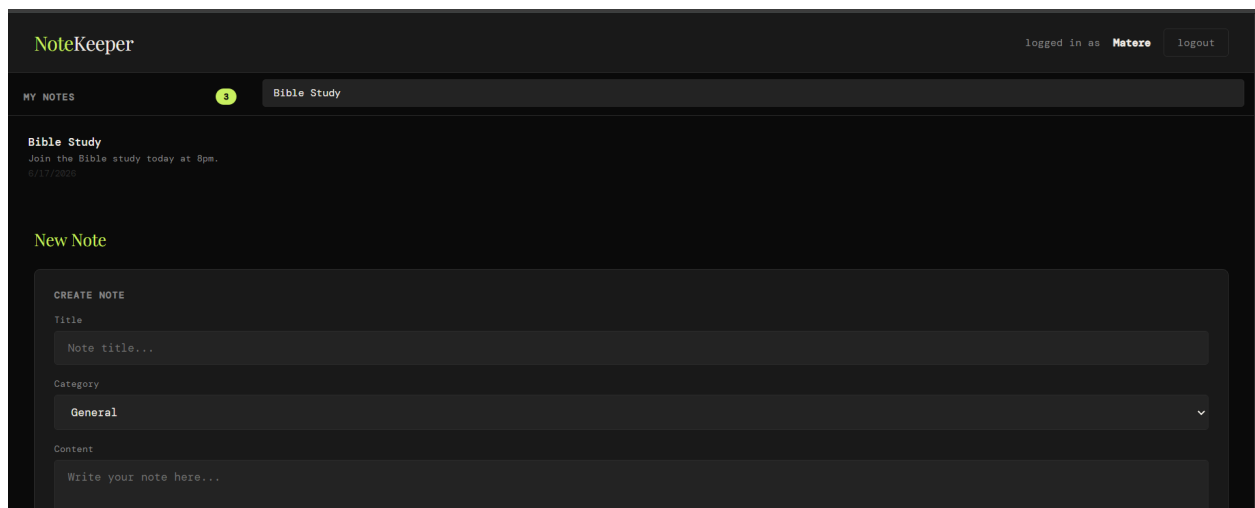


Fig 5: Edit Record

New Note

CREATE NOTE

Title

Tech Companies & Software

Category

Study

Content

Research about several tech companies in Nairobi.

Update Note

Fig 6: Delete Record

|                          |      | id                                   | user_id                              | title                     | content                                           | category | created_at          |
|--------------------------|------|--------------------------------------|--------------------------------------|---------------------------|---------------------------------------------------|----------|---------------------|
| <input type="checkbox"/> | Edit | 065373d4-ef12-453c-9a9a-43778fa81694 | 954c422b-6eb7-486f-92ed-99341205d3f2 | Bible Study               | Join the Bible study today at 8pm.                | Personal | 2026-06-17 08:29:58 |
| <input type="checkbox"/> | Edit | 4e52ad21-953c-494e-8829-fae1206837c6 | 954c422b-6eb7-486f-92ed-99341205d3f2 | Tech Companies & Software | Research about several tech companies in Nairobi. | Work     | 2026-06-17 08:29:36 |
| <input type="checkbox"/> | Edit | 82077cd8-5ae5-4865-83ec-692f802c00f3 | 954c422b-6eb7-486f-92ed-99341205d3f2 | Penetration Testing       | Watch tutorials on how to Pen Test.               | Study    | 2026-06-17 08:30:30 |

☐ Check all With selected: Edit Copy Delete Export

Fig 7: Responsive Design

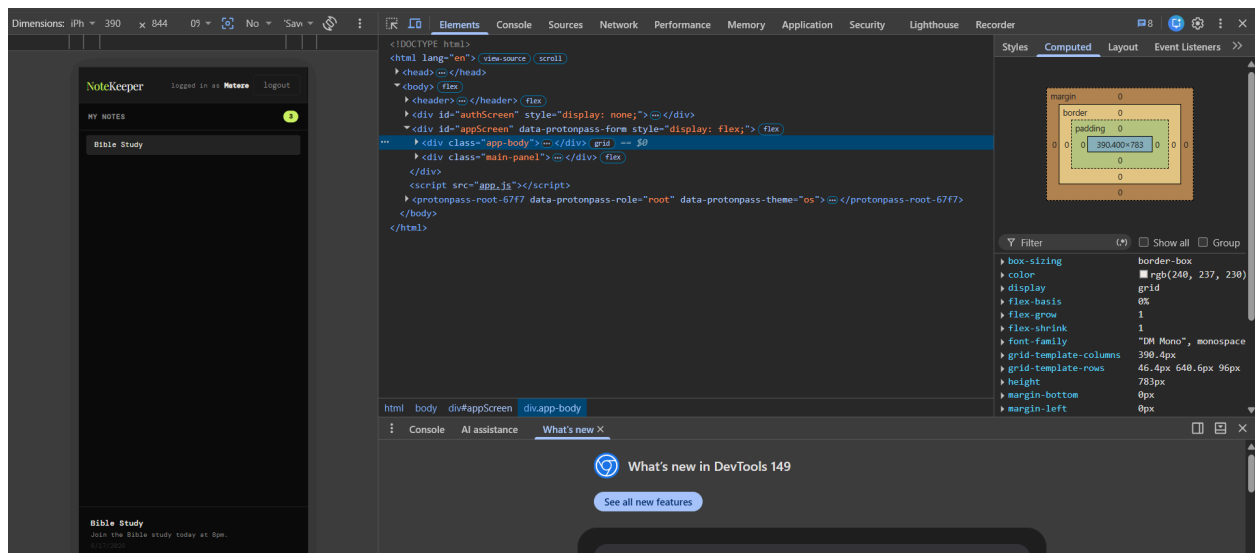




Fig 8: Form Validation

**Welcome back.**

Your notes, stored securely in MySQL.

[Login](#) [Register](#)

Username

choose a username

Email

you@example.com

Password

.....

Create Account →

Please fill in all fields